

Group 6

DATA 6550 Visualization Project

Topic: Good and Bad Visualizations of the Titanic Dataset

Instructor: Dr. John Wallin

Dataset Link: <https://www.kaggle.com/c/titanic/data>

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AI USAGE STATEMENT:

AI assisted tools helped us to structure and integrate the contributions of all group members into a comprehensive report. We also used AI to summarize the course readings and clarify the expression for consistency and clarity. Data analysis, visuals, and interpretations were done by the group members, and the content was thoroughly checked and approved by the team.

Introduction

The Titanic dataset contains statistical and survival data for 891 travelers who were on board the RMS Titanic amid its lady voyage in 1912. The dataset incorporates 12 factors portraying traveler personality, financial status, travel conditions, and survival results: PassengerId, Survived, Pclass, Title, Sex, Age, SibSp, Dry, Ticket, Passage, Cabin, and Set out. These factors capture statistical characteristics, family connections, and markers of financial status that permit investigation of survival designs amid the disaster.

The sinking of the Titanic speaks to a chronicled catastrophe including critical misfortune of human life. Hence, analyzing this dataset isn't as if it were a specialized work out but too an moral obligation. Visualizations made from such information must communicate designs precisely whereas dodging deceiving visuals that misrepresent or mutilate chronicled substances. This report analyzes how visualization choices can impact understanding and recognition of real-world occasions.

Dataset Overview

The dataset contains both numerical and categorical variables describing passengers:

1. Numerical Variables are:

- Age
 - Average Age: 29.70
 - Max Age: 80.00
 - Min Age: 0.42
- Fare
 - Average Fare: 32.20
 - Max Fare: 512.33
 - Min Fare: 0.00 (likely passengers working on the ship)
- SibSp (number of siblings/spouses)
 - Average SibSp: 0.52
- Parch (number of parents/kids)
 - Average Parch : 0.38
- Passenger Class (Pclass)
 - Median Pclass: 3.0 (indicating most passengers traveled in third class)

2. Categorical Variables are:

- Sex
 - Male: 577 (64.8%)
 - Female: 314 (35.2%)
- Embarked

- Cabin
- Ticket
- Name (provide additional contextual and categorical information)

3. Survival Outcome:

The target variable for this analysis is 'Survived', where:

- 1 indicates survival
- 0 indicates non-survival

The overall survival rate in the dataset is approximately 38.38%, meaning that approximately 342 of 891 passengers survived while the other 549 did not survive. The purpose of this analysis was to explore survival patterns across different demographic groups. Sibling/spouse and parent/children suggested that many passengers traveled alone or up to one person. Lastly, the dataset is mostly male at 64.8%.

From the descriptive statistics, most passengers were traveling in third class, as indicated by the median passenger class of 3.0. The average passenger age was approximately 29.7 years, ranging from infants to elderly individuals. Fare values varied widely, reflecting strong socioeconomic differences among passengers.

Missing Data Considerations

Some variables contain substantial missing values:

- Cabin is missing in approximately 77% of the observations (687 missing values)
- Age is missing in roughly 20% of the observations (177 missing values)

Missing data can skew data returns and can cause misrepresentation of results. In this dataset, Age and Cabin can cause skewness in subgroup comparisons due to the imbalance in the data.

Ethical Purpose of the Analysis

The purpose of this project is twofold:

- To make moral visualizations that precisely communicate patterns within the information.
- To intentionally develop deceiving visualizations illustrating how design choices, not data manipulation, can misshape interpretation.

By comparing moral and deceiving representations utilizing the same dataset, this extension highlights how visualization plan impacts understanding, fortifies the significance of mindful information communication, and interfaces analytical work to broader moral considerations in data science.

Dataset Loading and Preparation

The data on the Titanic was sourced from the Kaggle Titanic competition dataset and was brought into the analytical environment via Python. The dataset was converted into clean dataframes giving room for thorough investigations, plotting, and breakdown of survival results according to the diverse passenger attributes. For the analyses, the main dataset was the training set because it has the survival outcomes that were recorded, necessary for drawing the graphs and understanding the situation.

In order to create accurate visual representations, the dataset was carefully checked first to confirm that it would be consistent and useful. Information about data types, missing values, and column structures was gathered to be later used in understanding possible weaknesses, which would affect the interpretation of the data. Some variables like Age and Cabin had missing data points that were taken into consideration during the analyses to prevent the drawing of false conclusions.

Some basic data preprocessing steps were performed to ready the dataset for visualization. Among these actions were categorizing categorical variables, checking the numerical ranges, and generating new variables such as family size or fare classes to better identify passenger groupings. The derived features helped reveal the different characteristics of passenger groups without interfering with the original survival outcomes or changing the data set values.

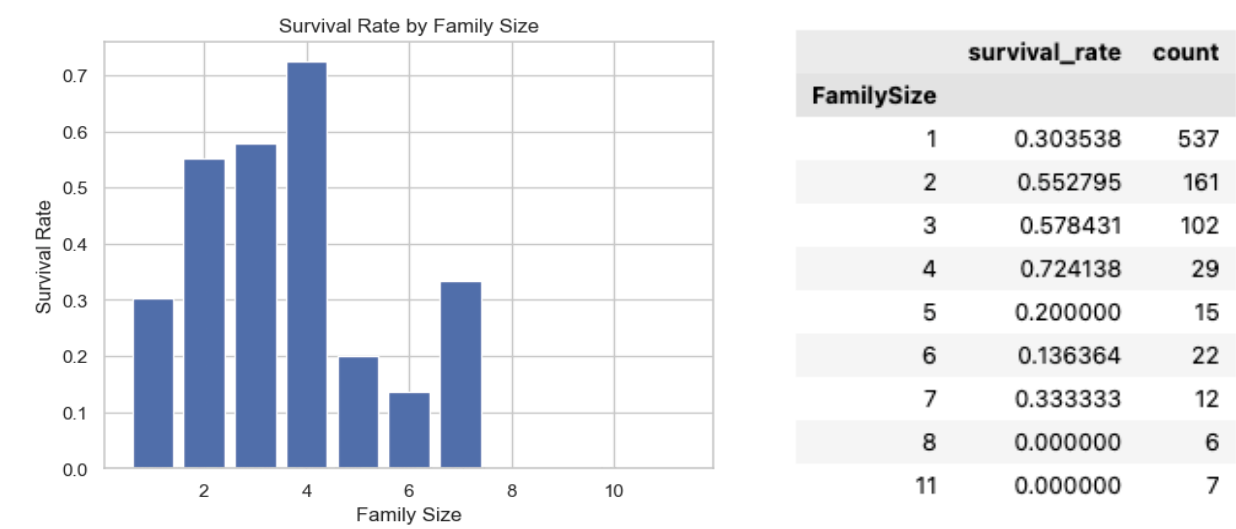
Ethically, it was crucial that the researchers always kept the data faithful to the true account of events when they were doing the preprocessing. They didn't take out any records nor did they manipulate any in such a way that the survival pattern would be artificially changed. The goal of preparation was solely to improve clarity and interpretability while preserving the authenticity of the historical data.

Exploratory Data Analysis (EDA):

An Exploratory Data Analysis (EDA) of the Titanic data was performed to gain insights into the data's structure and to figure out the most relevant passenger features that affected the survival outcomes. This step focused on the interpretative aspect, exploring, patterning, and identifying trends that could serve as guiding principles for ethically and visually effective representations, rather than on predictive modeling.

The first round of investigation looked at demographic and family variables as historical records reveal that factors such as travel conditions and social groups might have had an impact on survival in the disaster. An additional feature, FamilySize, was derived from the sum of siblings/spouses (SibSp) and parents/children (Parch) accompanying each

passenger. This variable measured whether a passenger was traveling alone or with a family and thus made it easier to correlate survival rate differences in divergent social environments.



The analysis illustrates survival chances depend quite a bit on the number of family members. It turns out that passengers in small families (2, 4 members) were most likely to survive, whereas solo travelers had a lower chance of survival. Very large families have much worse survival outcomes, though these categories are based on fewer data points. The observed trend strongly hints that moderate family support might have given some advantages, such as jointly deciding or helping each other in the evacuation situation, while too many people in one group or the difficulty of coordinating might have negatively affected larger groups.

Separately, the summary table that comes with the text supports these claims showing the survival rates and the number of passengers in each family size category. When the data are presented both visually and with numeric summaries, it provides a level of transparency as it gives anyone the opportunity to judge whether the trends are truly backed by data and not just by the look of the graph.

From a viewpoint of ethically sound visualization, this step of exploration was crucial as it later influenced design decisions without changing or selectively filtering the data set. The work is based on aggregated data obtained directly from the original records, thus the authenticity of the historical data has been maintained. The EDA process that involved a detailed study of patterns prior to the production of the final charts, on the one hand, avoided drawing misleading conclusions and, on the other, it made it possible to responsibly communicate insights based on a real, life tragic event.

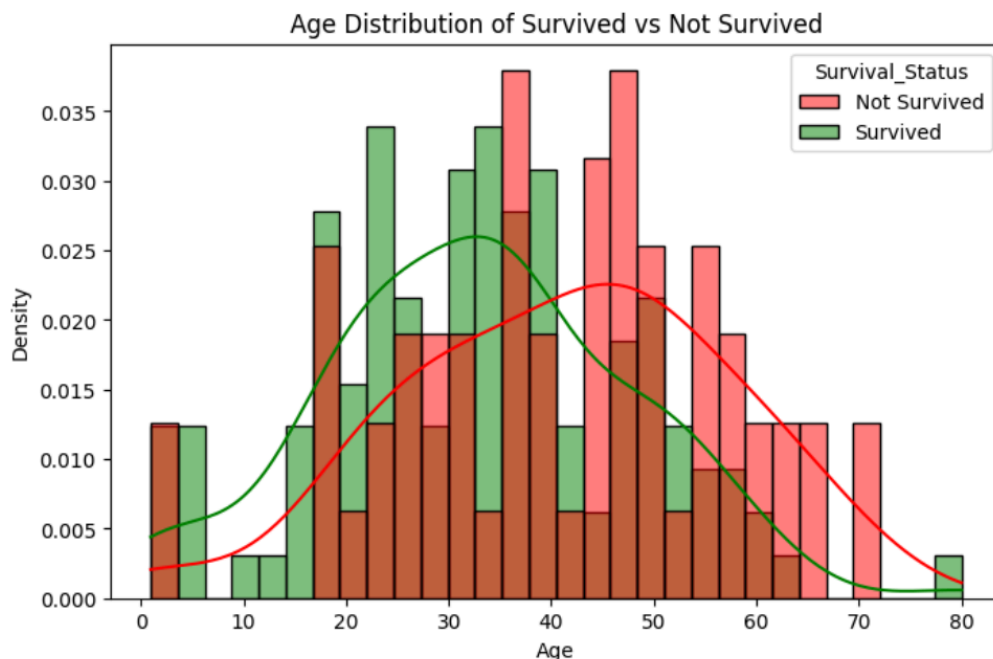
Ethical Visualizations

Ethical data visualization mainly focuses on accurately, transparently, and responsibly conveying the information without distorting the interpretation. Here, in the case of the Titanic dataset, the ethical visualization is very crucial as the data is about the real historical tragedy when many people lost their lives. Thus the graphic representations should be the first in the line for clarity, honesty, and contextual understanding rather than just the visual exaggeration or manipulation of emotions.

The below visualizations were made to show the survival trend using the right kind of charts, keeping the same scale, and neutral visual encoding. The data figures were not changed, partly filtered, or exaggerated in making the visualizations. Rather, aggregation methods like averages and proportions were applied directly to the original dataset to ensure a faithful representation of the historical outcomes.

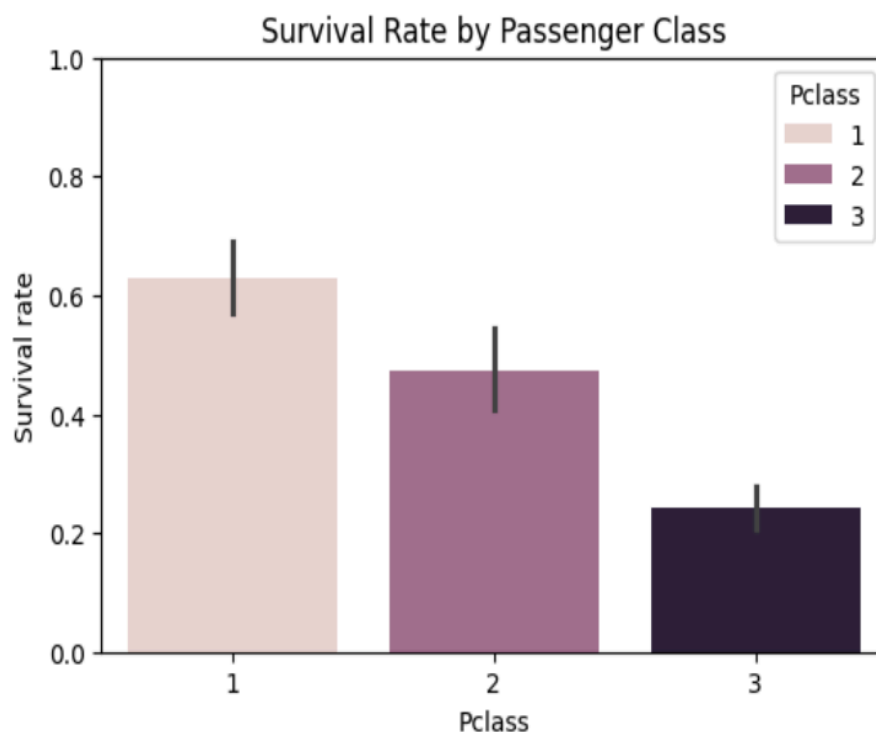
High emphasis has been placed on readability, accurate comparison, and proper labeling in each visualization so that the viewers can understand the data relationships without any confusion. Design decisions such as starting axes at zero, using proportional comparisons, and consistent application of color schemes not only prevent the creation of misleading impressions but also help to derive meaningful insights. These visuals are showing that responsible visualization skills can effectively communicate the patterns without compromising on ethics in data storytelling.

1. Age Distribution of Survived vs. Not Survived



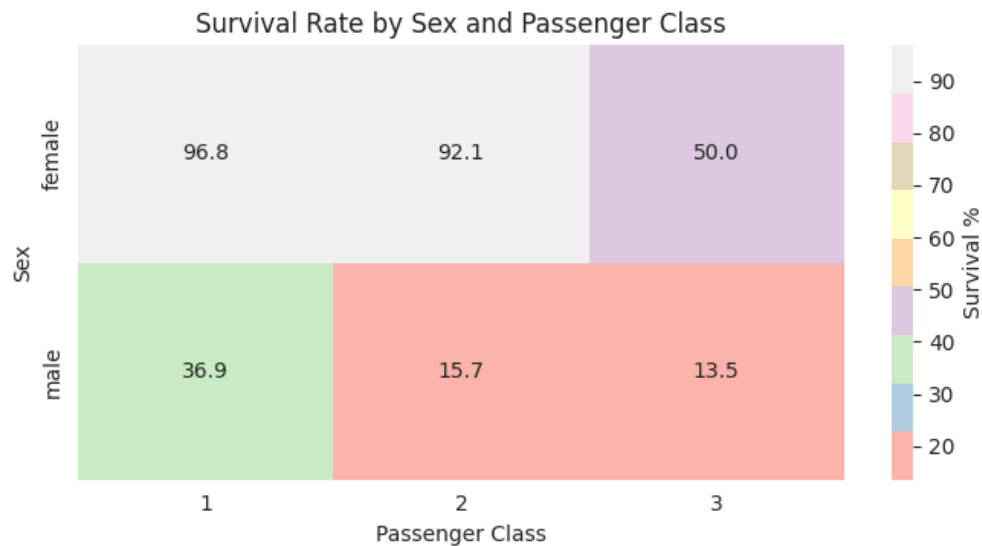
This visualization represents a density-based histogram comparing age distribution of passengers who survived vs those who did not survive. The histogram is looking at overall shape rather than raw counts. The choice in normalizing the distribution allows for a fair comparison of age patterns. Plotting raw counts could exaggerate differences due to age group sizes. The graph also uses green to show survived and red to show not-survived, which can help with interpretation of the graph. The graph is a good example of an ethically considered visualization.

2. Survival Rate by Passengers



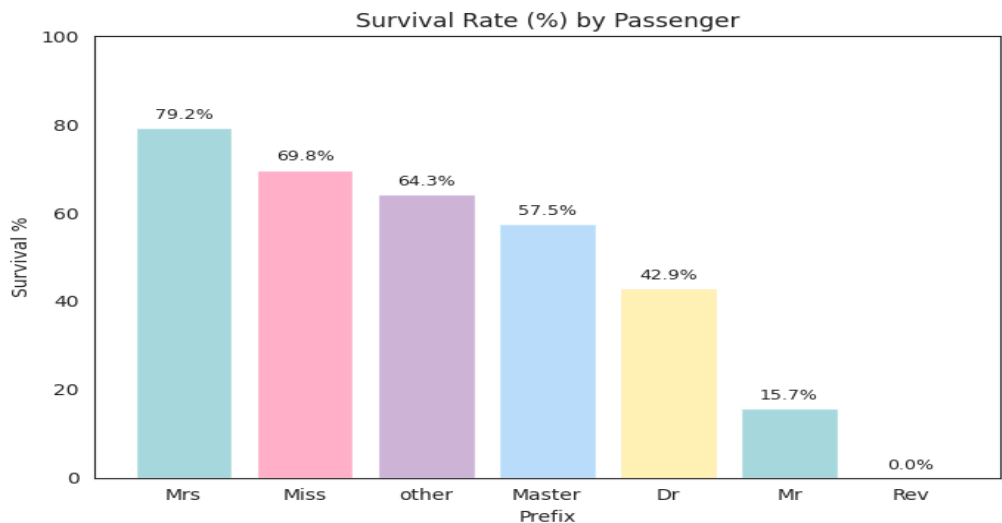
This chart is a representation of the connection between passenger class and the chance of survival. Survival rates depict a pattern of decreasing class of passengers from 1 to 3. The highest survival rate was for first class passengers, who were followed by second, class and third class passengers. This declining trend reveals that socioeconomic factors had a major impact on survival outcomes. The image depicts the disparity among passengers during the Titanic disaster. This chart is an illustration of an effective visual with insightful interpretation of the data.

3. Survival Rate by Sex and Passenger Class



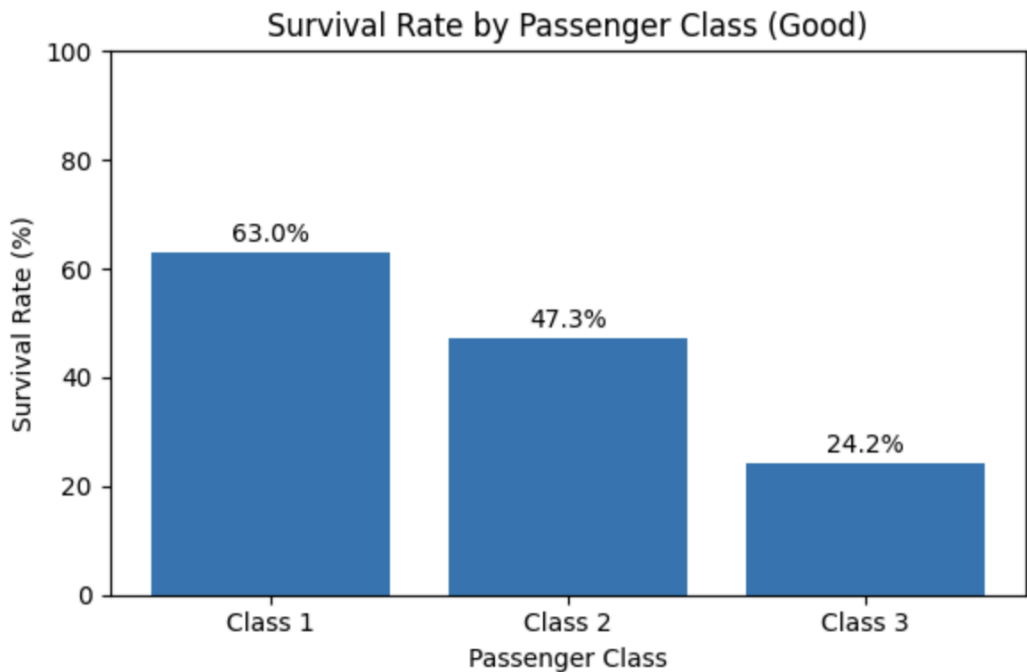
This graph compares the age distribution of passengers who survived and those who did not survive. Instead of just showing counts, it shows the overall shape of the age distribution. This helps us understand how age might have influenced survival. From the graph, we can notice that younger passengers had slightly better chances of survival compared to older passengers. The chart does not exaggerate anything. It simply shows the pattern clearly. Since it uses density instead of raw numbers, it allows a fair comparison between the two groups. This is a good visualization because it helps us understand the data without misleading the viewer. The colors are also clearly separated, so it is easy to read.

4. Survival Rate(%) by Passenger



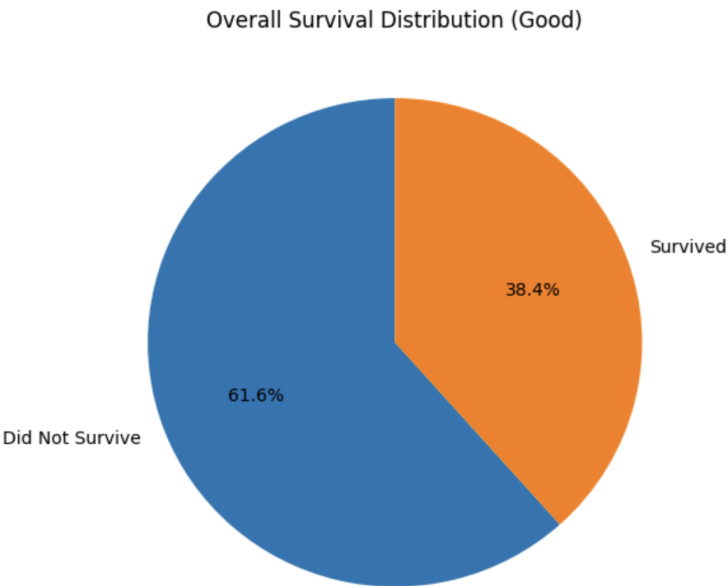
This chart shows survival percentages for first, second, and third class passengers. We can clearly see that first class passengers had the highest survival rate, and third class passengers had the lowest. This tells us that socioeconomic status played an important role during the Titanic disaster. The graph starts from zero and uses proper scale, so it does not distort the differences. It directly communicates the inequality that existed on the ship. This is an example of a strong and meaningful visualization.

5. Survival Percentage by Passenger Class



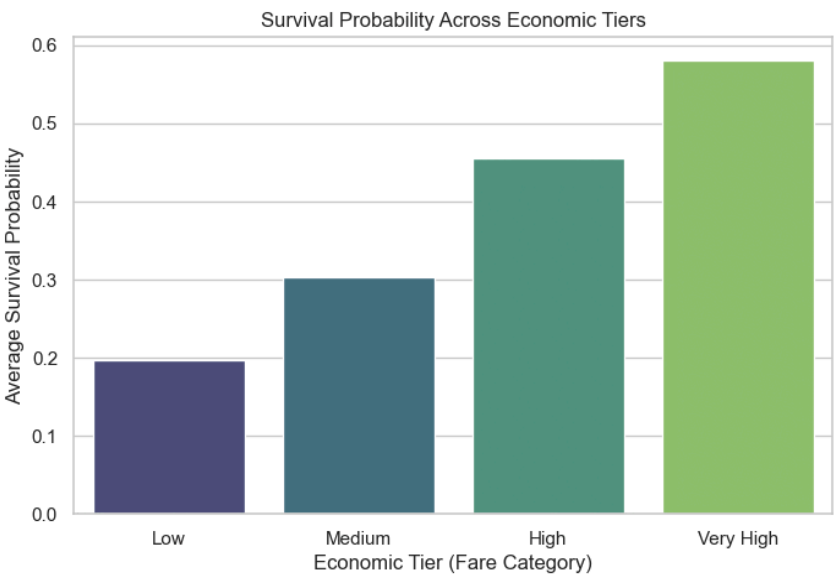
This bar graph illustrates the survival rate of the passengers based on the three categories of tickets on the Titanic. The findings clearly show that First Class passengers had the highest survival rate of approximately 63%, followed by Second Class passengers with a survival rate of approximately 47%, while Third Class passengers recorded the lowest survival rate of approximately 24%. This implies that the class of passengers significantly influenced the survival outcomes. The high survival rate among First Class passengers might be attributed to their proximity to life-saving boats, priority in evacuation, or advantageous positions of their cabins. On the other hand, Third Class passengers might have encountered more difficulties during the emergency.

6. Overall Survival Distribution



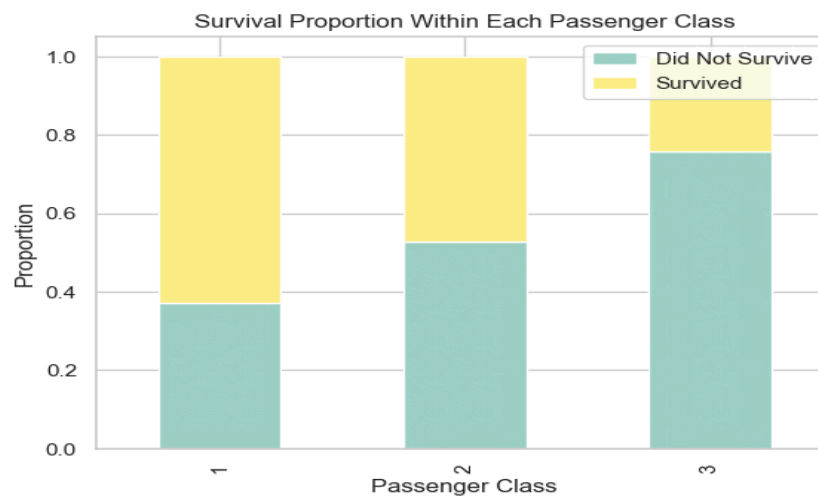
This pie chart represents the total survival distribution of the passengers in the Titanic. This chart shows that about 38.4% of the passengers were able to survive, whereas about 61.6% of the passengers were unable to survive. This chart clearly represents that a major portion of the passengers lost their lives in the tragedy. This chart is able to provide a correct and truthful representation of the survival results by showing the entire data without any filtration. The comparison between the two results is easy to recognize due to the percentage representation in the chart.

7. Survival Probability Across Economic Tiers



The chart clearly shows that survival probability increases gradually as the fare increases, indicating a positive trend, allowing the viewers to easily observe the relationship between economic and survival outcomes. The bar chart uses a consistent scale starting from zero enabling accurate comparison across fare categories without overhighlighting differences. The axis labels and the titles are clear and neutral helping the viewers to interpret the data easily. The passengers in the higher fare categories appear to have better survival rates which may be associated with cabin placement, accessibility or evacuation priority during the emergency event. On the other hand, the lower fare categories passengers tend to have lower survival rates than higher ones, which might be because they might have not given priority to evacuate due to their financial background.

8. Survival Proportion with each Passenger Class



The stacked bar chart clearly shows the survival proportion varies across passenger classes, and it allows the viewers to observe how the survival outcomes changed, depending upon the passenger's travel class. The bar chart uses proportional representation which in turn helps comparing the survivors and the non-survivors fairly and clearly within each class instead of just focusing on the total counts of the death. The scale starts from zero and the labels, legends are clearly defined, which enables the viewers to interpret the data on the stacked chart without any confusion. Passengers that are traveling in the first class appear to have higher survival rates as compared to those who are in third and second class which will be associated with the cabin location of the passengers, accessibility to the lifeboats or even the evacuation priority during an emergency situation. And on the other hand, passengers in the lower classes show lower survival proportions, which clearly suggests that class factors may have influenced the survival chances of the passengers while the visual presents this relationship clearly without exaggerating or manipulating the data representation.

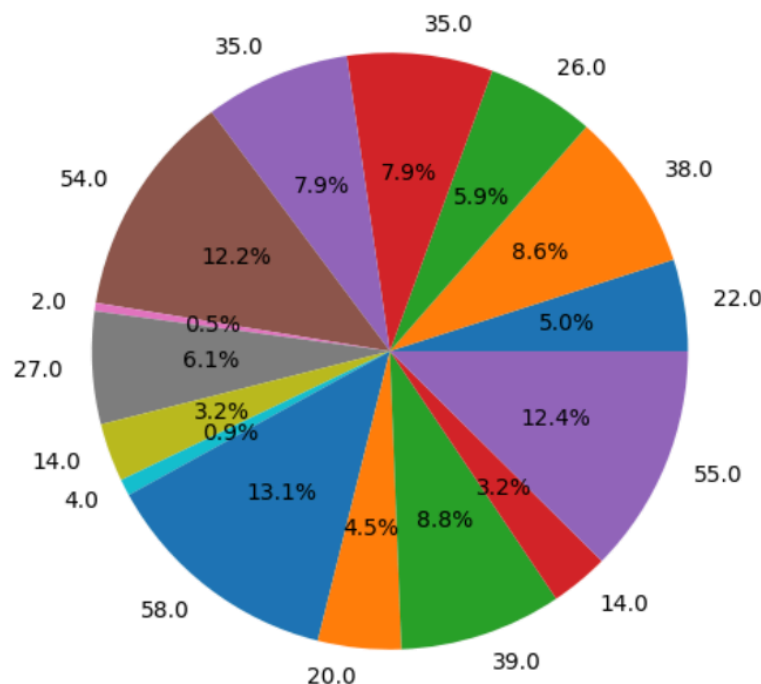
Intentionally Misleading Visualizations

As a part of ethical visualization, the goal is generally to communicate insights clearly and responsibly, however, one can still get a distorted perception due to the designer's choices in the visualization. This section is about how visual encodings, scales, stylistic elements, and presentation choices can change the way people interpret information.

Four misleading visualizations can be traced back to the same Titanic dataset used without changing or fabricating any data values. Instead of adding misleading effects through the use of visualization design decisions such as truncated axes, exaggerated styling, unnecessary visual complexity, and emotionally suggestive formatting, these products are highlighting the fact that despite the data being valid, the viewers will arrive at wrong or overstated conclusions.

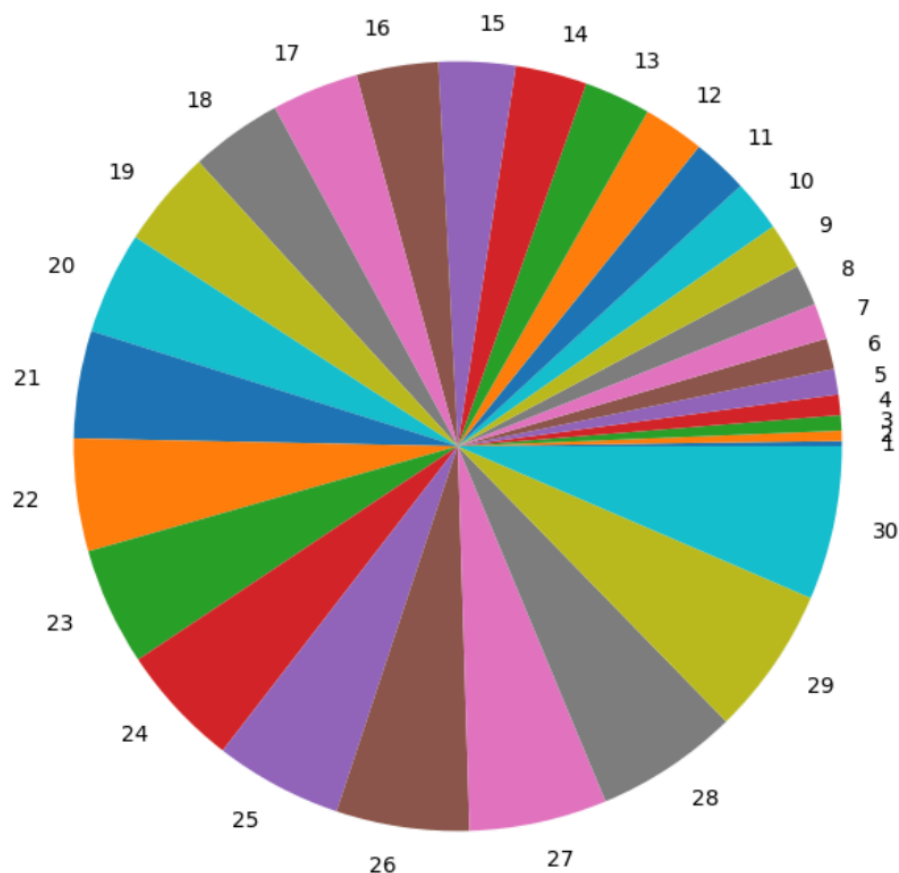
Since the Titanic dataset is an account of a real life tragedy, the misleading visualizations serve to highlight the important ethical issue in data science, that poor visualization practices may unintentionally shape stories, reinforce existing bias, or trivialize complex events. This is further underlined by the responsibility of the analysts to communicate data truthfully and in a transparent manner as evidenced through the comparison of the visuals with the ethical versions shown earlier.

1. Using Unrelated Plots - 1



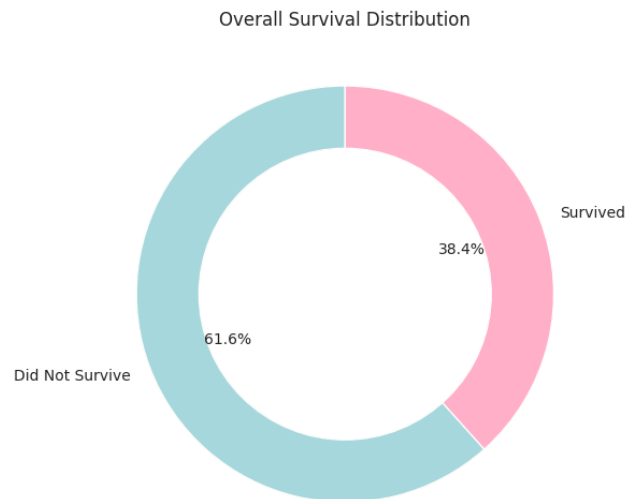
This pie chart attempts to display individual passenger ages as a proportional slice. Each slice represents a specific age from the passenger dataset. This chart is technically correct, but doesn't contain meaningful data and doesn't use the correct data choices when using a pie chart. This chart is confusing to look at, too. There isn't an appropriate title, it's treating age as proportional categories, and it's very busy. Pie charts are designed to represent categorical proportions that sum to a whole (100%). Age doesn't meet this criteria. This visual is an example of a bad visual design as it can distort someone's understanding of the data.

2. Using Unrelated Plots - 2



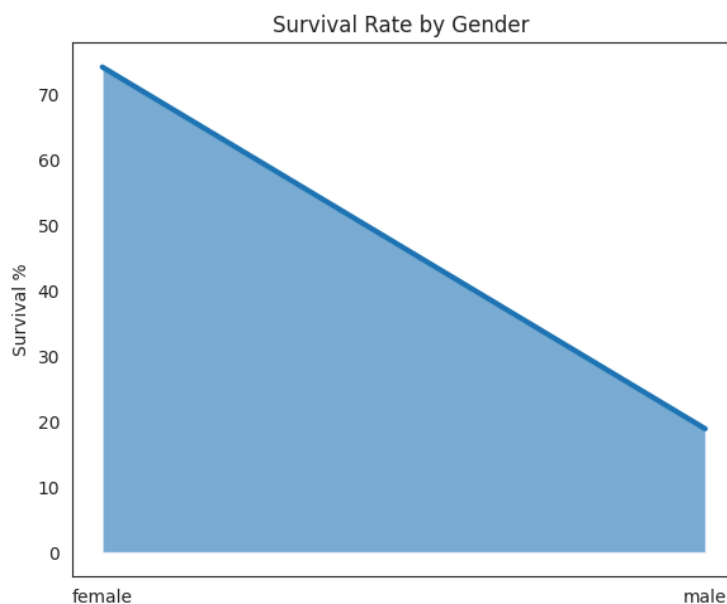
This pie chart is representing passenger IDs as slices of a whole. Passenger ID is a unique identifier, so it doesn't have analytical meaning. This chart is very busy, and offers no meaningful interpretation. Someone viewing this chart could possibly assume that there could be some type of relationship since the slices are relatively thin, however, this can lead to distrust in a dataset if a chart like this is included in an analysis of the dataset. It seems to appear like something substantive, but is analytically meaningless.

3. Overall Survival Distribution



The chart shows the overall percentage of passengers who survived and who did not survive. While the chart is not technically wrong, it does not provide much insight. It only shows two categories and does not explore patterns like gender, age, or passenger class. A simple bar chart could show the same information more clearly and accurately. The chart looks visually good, but it does not add strong analytical value. It focuses only on the final outcome and ignores the factors that influenced survival. This is an example of a visualization that is simple but not very informative.

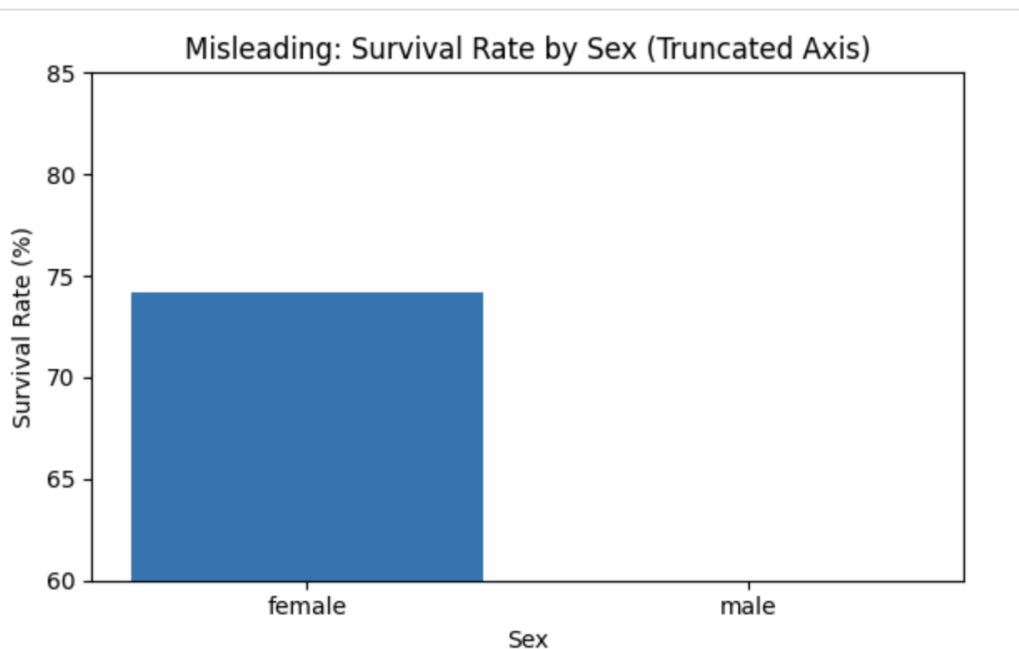
4. Survival Rate by Gender



This chart shows survival rate for males and females using an area style graph. Even though the values are correct, the area design is not the best choice for this type of data. Area charts are usually used to show trends over time. In this case, gender is not a continuous variable or time-based variable. It only has two categories: male and female. Because of this, the filled area makes it look like there is gradual change, which is not true. The large shaded area can also exaggerate the difference between male and female survival rates. It makes the difference appear bigger than it actually is. A simple bar chart would tell the difference more clearly and accurately.

So while the data itself is correct, the choice of visualization type makes this chart less appropriate.

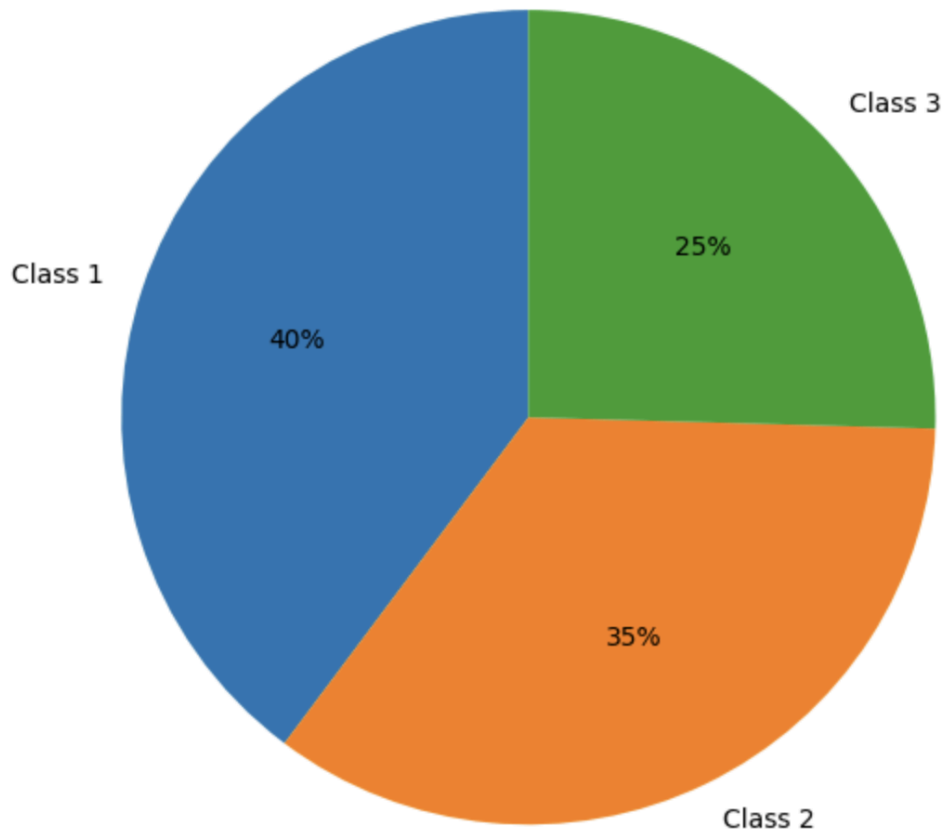
5. Survival Rate by Sex (Truncated Axis)



The bar graph shows the survival rate based on gender, but the y-axis has been set to a false scale to emphasize the gap between the survival rates of men and women. The y-axis starts from 60%, rather than 0%, to make the gap between men's and women's survival rates appear larger. Although it is true that the survival rate of women is higher than that of men, the false scale of the y-axis gives a wrong impression of the gap. The data is still accurate, but the presentation of the data is false. This is an example of how a false axis can mislead the audience and give a wrong impression of the data.

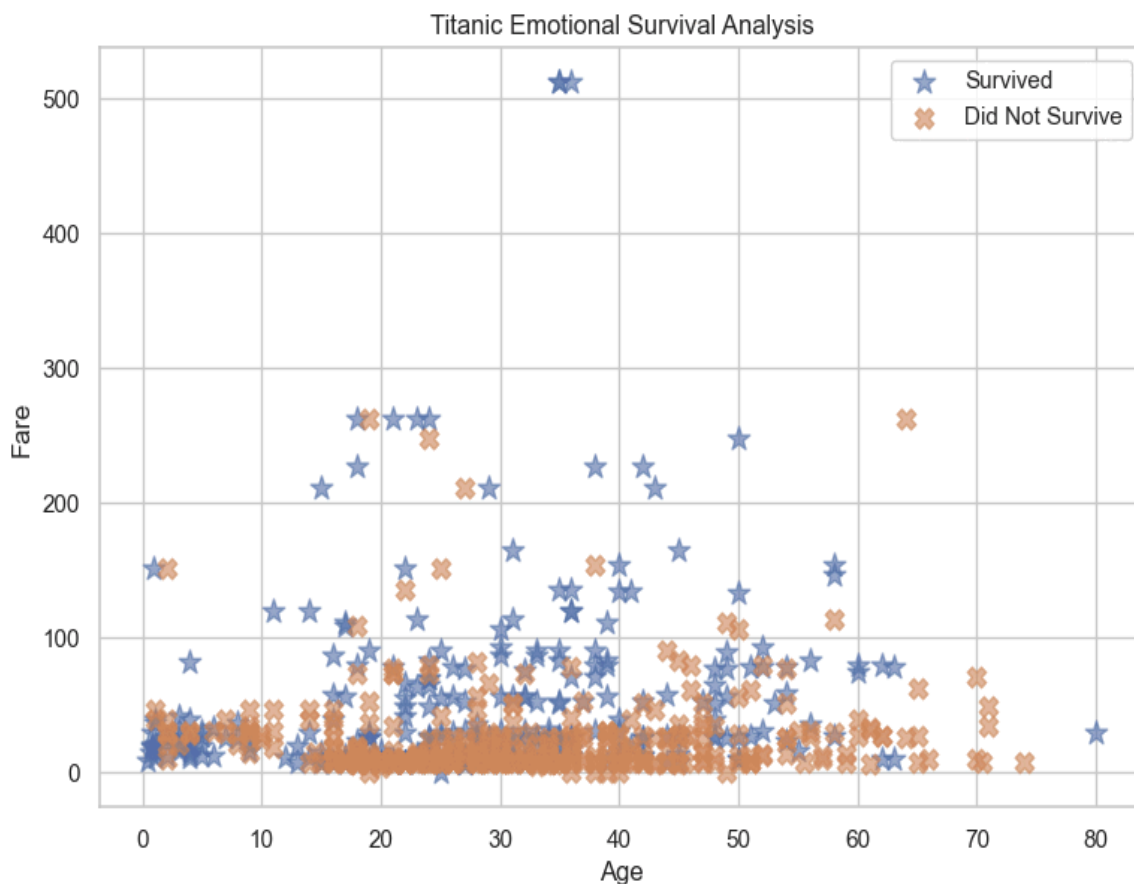
6. Class Distribution Among Survivors Only

Misleading: Class Distribution Among Survivors Only



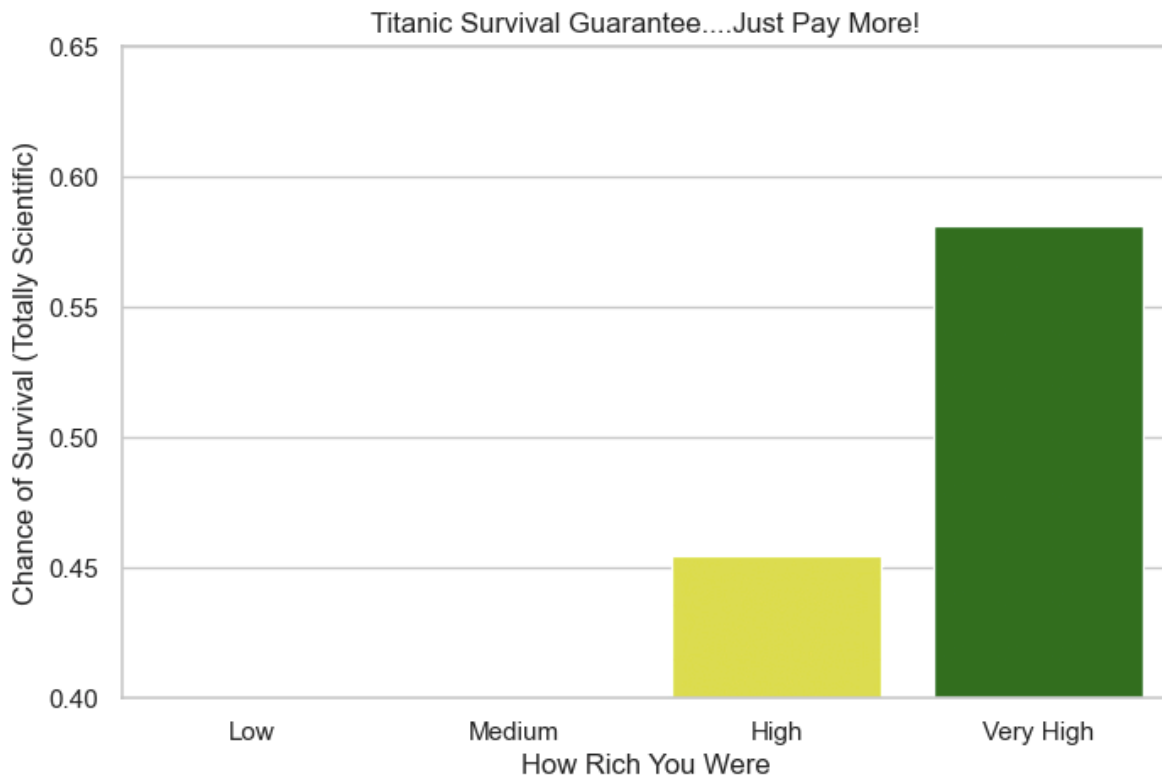
This pie chart represents the number of survivors of the Titanic shipwreck divided into classes of passengers. At first glance, it might seem that the majority of the survivors belonged to the First Class (40%), then to the Second Class (35%), and finally to the Third Class (25%). However, it is important to note that this visualization is misleading. This is because it does not represent the full picture of the number of people from different classes who were on the ship. Since the number of people in the First Class was smaller than in the Third Class, it might create an impression that the number of survivors from different classes was more or less equal. This is an example of selective filtering or "cherry-picking". Cherry-picking data is a common way visualizations can unintentionally or intentionally mislead an audience.

7. Emotional Survival Analysis



The scatter plot displays the relationship between passenger age and the fare that they paid with their survival outcomes. However, the use of the symbolic marker makes the visualization difficult to interpret clearly. Oversize stars and cross symbols overlap heavily in which they are hiding several data points and preventing the viewers from accurately identifying patterns or comparison within the points in the data set. The symbolic shapes introduce emotional meaning that distracts from analytical interpretation and shifts attention of the audience towards visual appearance rather than the data itself. Even though the data set remains unchanged throughout the scatter plot, the design choices reduce the clarity of the concept and create a confusion in the minds of the audience. This visualization is considered unethical because it emphasizes stylistic elements over readability and accurate communication which in turn shows how appropriate visual encoding can mislead viewers without altering the actual data.

8. Titanic Survival Sarcasm Plot



The bar chart appears to show that survival probability increases significantly as the fare category increases, which may lead the viewers to believe that paying more is almost guaranteed survival. This is a sarcasm graph where the visualization uses a truncated Y-axis that does not start from zero and which exaggerates the visual differences between the categories and makes the survival gap appear much larger than it actually is. The sarcastic title and emotionally suggestive axis labels further influence the interpretation by framing the results in a biased manner rather than presenting them neutral, although the underlying data values remain accurate and were not modified. The design choices manipulate the overall meaning in the perception the viewer takes towards a predetermined conclusion. This visualization becomes unethical because it prioritizes dramatic storytelling over objective communication and in turn, it demonstrates how scaling and wording decisions can distort the understanding even when the data itself underneath is correct.

Discussion and Ethical Reflection

This project proves that visualization goes beyond mere technical procedure and is also a kind of societal accountability. We worked with the Titanic dataset to demonstrate how the same data could lead to very different interpretations based on the visualization design decisions made.

Ethical visualizations which were discussed earlier have the goal of pattern communication through the use of suitable scales, unemotional styling, and clear labeling so that the viewer can correctly interpret the trends in survival without any exaggeration.

On the other hand, the misleading visualizations demonstrated that one does not need to alter the underlying data to mislead. The implementation of these misrepresentations can be done by the means of a truncated axis, the exaggeration of visual elements, the use of emotionally suggestive colors, the inclusion of unnecessary dimensions, or the creation of crowded representations. These design choices change the way in which the viewer sees the information and thus the viewer is very likely to make an exaggerated or incorrect conclusion about survival outcomes which is contrary to the fact that the numerical values were still correct.

The Titanic dataset is a record of a real historical tragedy that involved a great loss of human life. Given this, ethical visualization is of utmost importance. Showing the data irresponsibly might, on the one hand, fail to represent the complexity of human experiences and, on the other hand, may unintentionally lead to misleading narratives about socio, economic status, chances of survival, or blame for the disaster. Hence, visualization designers have a responsibility to strike a balance between clarity, accuracy, and respect for historical facts.

This comparison of ethical and misleading charts is a great example of a fundamental lesson of data science: charts and other visualizations are powerful tools of persuasion. People tend to trust a visual presentation more than a table of numbers, so a visually inadequate design could heavily skew the audience's understanding even in the case of accurate data. Following visualization ethics, honest scaling, selecting the right chart, and making clear labels are some of them, is the fine line between insight communication and information distortion.

In general, the work done here lays bare the dependence of ethical visualization on well, informed choices of analysts. They have to go beyond looking at the data in one dimension and think of how the means of presentation will determine the message received. Good, safe visualization goes hand in hand with openness, it facilitates general comprehension, and it is a source of confidence in data- driven communications.

Conclusion

This project is geared towards understanding how data visualization can help people to comprehend data better through analyzing the survival patterns of the Titanic dataset. The researchers conducted the exploratory data analysis and feature exploration and found out that passenger characteristics such as class, fare, family structure, etc. had a significant influence on survival outcomes.

They then created ethical visualizations which clearly communicated these findings through the use of appropriate scales, transparent labeling, and unbiased design choices.

On the other hand, intentionally misleading visualizations show how a person's interpretation can be distorted purely by the presentation decisions even if the underlying data are accurate.

They contend that by truncating axes, changing styling dramatically, and adding needless visual complexity, for instance, one can trick the audience to come to wrong conclusions without them realizing the role of the design in the decision.

By comparing the ethical and misleading confusions it has become clear that visualization is not only technical but also ethical. Data analysts have a responsibility to consider the way how their visual decisions may influence perceptions especially when the datasets represent real historical events and human experiences. Responsible data visualization ensures that we keep the data faithful, we open the data, and we can count on the data that is shared with us.

Ultimately this study confirmed the necessity of having strong analytical skills coupled with ethical consciousness. Visualizing effectively means not only analyzing correctly but also communicating sensitively with respect to both the data and the audience interpreting the data.